Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/02/2005
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Project Title:	Crystallization of the Mitochondrial Metabolite Transport Proteins		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60064-3095	**Congressional District:**	10
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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	The above objective will be accomplished via the following 4-step plan. First, the dispersity of purified preparations of the five mitochondrial metabolite transporters will be determined in a panel of detergents via dynamic light scattering. With preparations that are found to be monodisperse we will initiate systematic and comprehensive crystallization trials. With those transporter preparations that are found to be polydisperse, we will vary the detergent, as well as other constituents present in the purified transporter solution. Second, we will identify and optimize conditions that yield X-ray diffraction quality crystals of the transporter preparations by conducting screens utilizing a several-pronged approach consisting of: i) the use of a sparse matrix methodology; ii) the use of more narrow targeted screening employing conditions that have proven successful with other		

Task Description:	membrane proteins; and iii) identification of reagent combinations that yield second virial coefficient (i.e., B22) values within the "crystallization slot" followed by systematic crystallization trials utilizing these conditions. The third step will consist of the application of more novel crystallization strategies to the mitochondrial transporters in an attempt to improve crystal quality and size. These strategies will include antibody-assisted crystallization, the use of bicontinuous lipid cubic phase crystallization, and containerless crystallization in floating drops. The final step will involve the establishment of crystallographic parameters of crystals grown under terrestrial conditions for eventual comparison to crystals grown in a microgravity environment. Crystal characterization will include a detailed analysis of three-dimensional intensity X-ray diffraction data sets, as well as photodocumentation. The objective of this program is to identify and optimize conditions that will enable the growth of X-ray quality crystals of five mitochondrial metabolite transporters - the dicarboxylate, oxodicarboxylate 1, oxodicarboxylate 2, oxaloacetate, and succinate/fumarate transporters. Once diffraction quality crystals are obtained with a given transporter, we will initiate a series of control experiments designed to characterize crystals grown under terrestrial conditions. This will provide the foundation for the long-term goals of these studies which are to: i) compare the properties of crystals grown on Earth with those grown in the microgravity environment of space; and ii) solve the structure of one or more of these carriers at high resolution. The significance of the proposed research is several fold. First, the mitochondrial transporters play essential roles in cellular bioenergetics in both the normal and pathological states. Despite their importance there is a paucity of information regarding their structural properties, owing primarily to the difficulties in preparing well-diffracting crystals. Knowledge of the structure of these carriers is critical for structure-based drug design, as well as a basic understanding of their functioning in living cells. Second, it is thought that since common structural and functional principles likely apply to a diverse group of both mitochondrial and non-mitochondrial transport proteins, a high resolution structure of one or more mitochondrial metabolite transporters will have a significant impact not only on our understanding of these carriers, but also on a larger sphere encompassing diverse membrane transport processes. Third, the proposed research will have an important impact in the more general area of understanding the structural basis for membrane protein function. It is striking that even though approximately 30% of open reading frames in eukaryotic cells encode polytopic membrane proteins, and that this class of proteins is of particular biological and pharmacological importance, there are still very few examples of high resolution structures for members of this class. This lack of structural information arises from the difficulties inherent in obtaining large quantities of purified membrane protein (a difficulty which we have already surmounted), and in identifying conditions enabling the growth of X-ray quality crystals of this class of macromolecule. Finally, the intrinsic flexibility of transport proteins, together with the relatively nonspecific interactions that stabilize membrane proteins within the lipid bilayer, make it very difficult to obtain protein crystals that diffract to high resolution on Earth and thus these macromolecules are particularly attractive targets for investigation in the microgravity environment of space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the final year of this grant investigations were conducted in the following areas: 1) Use of the methodology that we developed to quantify detergent levels in preparations of purified, concentrated transport protein immediately prior to initiating crystallization trials, led to the alteration of our purification methodologies in order to modulate the final detergent micelle/transporter monomer ratio. Trials were set up at various ratios and significant effects on crystal growth and morphology was observed. However, well-diffracting transporter crystals were not obtained. 2) Crystallization trials were carried out in agarose gels within capillary tubes in order to minimize the effects of convection, with the aim of obtaining more optimized nucleation and crystal growth. While subtle effects on crystal shape were seen, diffraction did not improve. 3) We attempted to increase the mechanical stability of crystals by cross-linking with glutaraldehyde. Furthermore, hanging drop crystallization trials were conducted in the presence of different lipids. However, little improvement in crystal quality was noted. 4) Finally, we began testing the ability of monoclonal antibodies generated against one mitochondrial transporter to bind to other mitochondrial carriers, with the long-term aim of using the Fabs generated from one antibody in crystallization trials with multiple mitochondrial carriers. In summary, many approaches have been investigated during the past year and we have made progress in assessing which ones are likely to be of greater importance in the continued development of higher quality transporter crystals in the future.
Bibliography Type:	Description: (Last Updated: 04/09/2021)
Articles in Peer-reviewed Journals	Eriks LR, Mayor JA, Kaplan RS. "A strategy for identification and quantification of detergents frequently used in the purification of membrane proteins." Anal Biochem. 2003 Dec 15;323(2):234-41. https://doi.org/10.1016/j.ab.2003.09.002 ;PMID: 14656530 , Dec-2003